VIT® Vellore Institute of Technology (Deemed to be University under section 3 of UGC Act, 1956)

School of Electronics Engineering (SENSE)
B.Tech. Electronics and Communication Engineering (specialisation in Biomedical Engineering)
BECE497J Project-I · Final Review and Poster Presentation · 29 October 2026

Code, data and demo
github.com/lvxn404/semantic-decoding-tribe-inversion

Conditioning and Non-Convexity of Semantic Decoding by Inversion of Deep Brain Encoding Models

Predictable is not recoverable: inverting Meta's TRIBE v2 foundation brain encoder on synthetic and real fMRI

Team members and Guide

- **Abhranil Sarkar** (23BML0077) · computational track: model instrumentation, inversion engine, experiments, real-fMRI pipeline
- **Tanisha Choudhuri** (23BML0081) · neurobiological track: cortical networks and ROI definition, dataset curation, interpretation
- **Guide:** Dr. Sharmila Nageswaran, Associate Professor, School of Electronics Engineering (SENSE), VIT Vellore

Project Overview

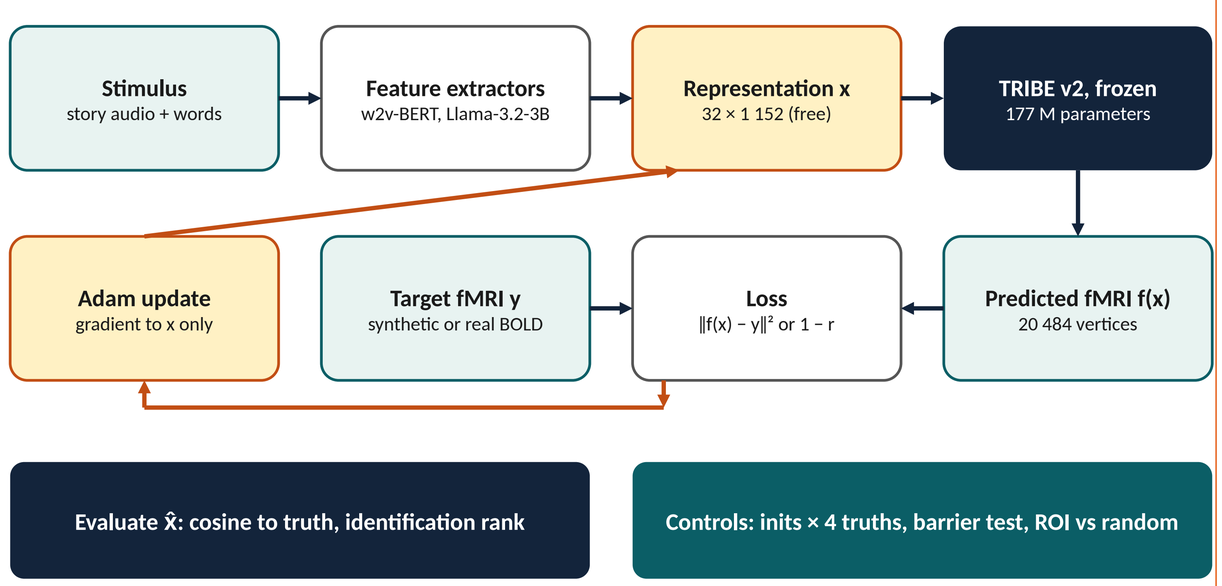
Brain encoding models predict fMRI from what a person sees and hears; decoding runs the other way. We ask whether a near-perfect encoder can simply be inverted to read meaning back out of the brain.

We invert Meta's open 177 M-parameter TRIBE v2 encoder by gradient descent, measure **uniqueness, conditioning and regional recoverability** with exact ground truth, and validate on real fMRI from a held-out story.

Justification of Complex Engineering Problem

- **No closed-form answer.** The forward model is a 177 M-parameter nonlinear transformer mapping a 32 × 1 152 representation to 20 484 cortical vertices; its inverse cannot be written down and must be measured.
- **Ill-posed by nature.** Existence, uniqueness and stability of the inverse (Hadamard) are all open: fMRI is slow, blurred and noisy.
- **No ground truth in real data.** The true meaning behind a brain recording is unknown, so we built a synthetic framework with exact error, then a real-fMRI test with identification at a known chance level.
- **Engineering obstacle.** The released pipeline is not differentiable (speech synthesis, transcription, frozen extractors); we found and re-entered the one differentiable boundary.
- **Conflicting requirements.** Decoding fidelity vs non-invasive, low-cost acquisition vs mental privacy cannot all be maximised.
- **Interdisciplinary.** Inverse-problem theory, deep learning, neuroimaging (surface registration, atlases) and neuroanatomy must work together.

Proposed Block Diagram



Only x is optimised; network weights stay frozen. Re-entering after feature aggregation reproduces TRIBE v2's predictions exactly (r = 1.0000).

Design Methodology and Novelty

1 Forward operator. TRIBE v2 (trained on 451.6 h of fMRI from 25 people watching films and video clips and listening to stories), frozen.

2 Inversion. $\hat{x} = \text{argmin} \|f(x) - y\|^2$, Adam (lr 0.05, cosine schedule), weights frozen.

3 Exact evaluation. Known synthetic x → y = f(x) → recover x; budget, noise, extent, priors, 8 initialisations × 4 truths.

4 Mechanism tests. Forward-only barrier test along solution differences; Jacobian sensitivity vs 200 random directions.

5 Regions and real data. 6 Destrieux networks vs size-matched random sets; LeBel 2023 fMRI projected to fsaverage5, held-out story.

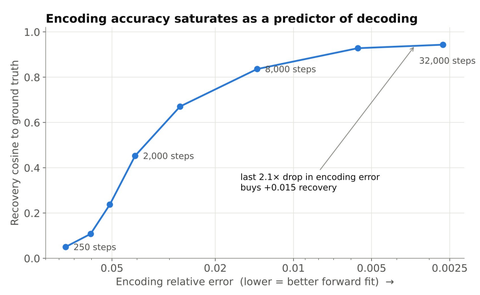
Why fMRI, not EEG: millimetre detail on 20 484 vertices, the only deep encoder large enough to invert, and one inverse problem instead of two (EEG adds source localisation). EEG is faster and cheaper, which matters for a device, not for this measurement.

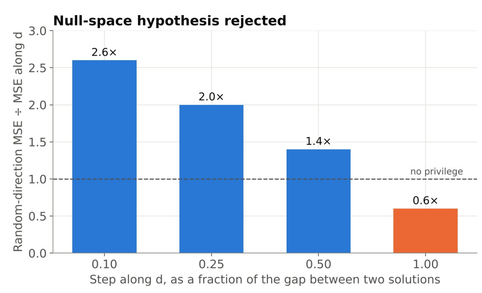
Novelty: closest prior work	Deep encoder	Optimises input	Tests uniqueness	Per region
Kay et al. 2008	○	○	○	○
Gardner & Liu 2019 (linear IEM)	○	●	●	○
Tang et al. 2023	○	●	○	●
TRIBE v2 (2026) and its two follow-ups	●	○ / ●	○	○
This work	●	●	●	●

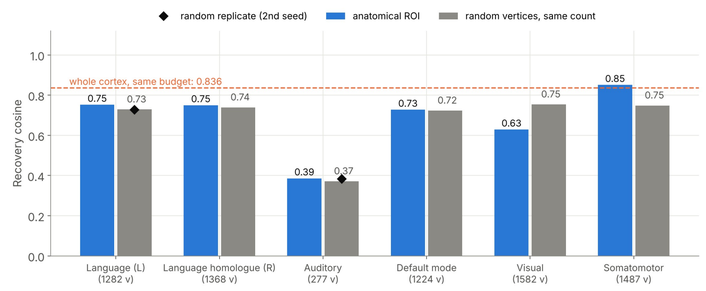
Literature search (9 Oct 2026, arXiv / Google Scholar / web): no published uniqueness or conditioning analysis of a deep foundation brain encoder.

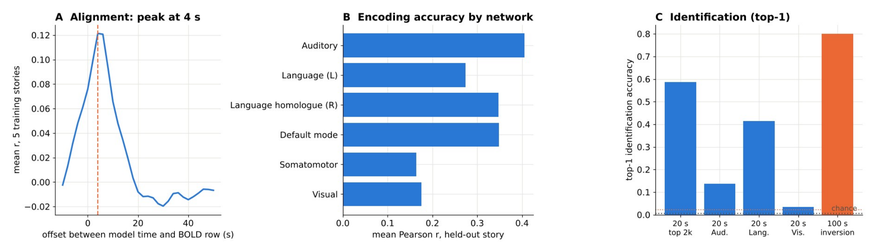
Project Results

A good fit is not a good decode, and the inverse identifies but does not reconstruct.









Synthetic: error ↓28.4× but recovery saturates (0.944); equal fits disagree (cosine 0.817); barrier, not null space. **Regions:** visual −0.125, somatomotor +0.102 vs random. **Real fMRI:** r = 0.48 (56% of ceiling); 4/5 segments identified (chance 2.2%), cosine 0.02.

PO and PSO mapping

PO1 Linear algebra, optimisation and inverse-problem theory applied to a neural operator.

PO2 Formulated decoding as an ill-posed inverse problem and analysed its causes.

PO4 Designed controlled experiments with exact ground truth and matched controls.

PO5 PyTorch, TRIBE v2, ANTs, nilearn, GPU (Colab) computing.

PO6, PO8 Mental-privacy and clinical-use implications; open data, licences respected.

PO9, PO10 Two-track teamwork; IEEE manuscript, poster, reviews.

PO12 Self-taught foundation models and neuroimaging tools.

PSO2, PSO3 Analysis and imaging of biosignals (fMRI) with modern tools.

WP mapping

WP1 Depth of knowledge: neuroscience, fMRI, deep learning and inverse-problem theory together.

WP2 Conflicting requirements: fidelity vs non-invasiveness vs privacy.

WP3 Depth of analysis: separates non-convexity from information loss with falsifiable tests.

WP4 Familiarity of issues: slow, noisy, blurred BOLD; no ground truth in real data.

WP7 Interdependence: ROI anatomy and computation depend on each other.

SDG mapping

SDG 3 (Target 3.4): well-being for people with ALS, stroke or locked-in syndrome who keep comprehension but lose speech: we measure what non-invasive decoding can deliver.

SDG 9 (Target 9.5): open, reproducible research capability in neural-interface engineering on free compute.

SDG 10 (Target 10.2): communication access is a precondition for inclusion of people with severe motor disability.

References

[1] S. d'Ascoli et al., TRIBE v2, arXiv:2605.04326, 2026. [2] J. L. Gardner, T. Liu, eNeuro 6(2), 2019. [3] J. Tang et al., Nat. Neurosci. 26, 2023. [4] A. LeBel et al., Sci. Data 10, 555, 2023. [5] K. N. Kay et al., Nature 452, 2008. [6] P. C. Hansen, Discrete Inverse Problems, SIAM, 2010. [7] J. Lévy et al., arXiv:2502.17480, 2025. [8] C. Destrieux et al., NeuroImage 53, 2010.

Project outcome

Complete and reproducible: IEEE-format manuscript (ready for submission), open GitHub repository with code, results, demo video and Colab notebooks, review decks and this poster. Finding: encoding accuracy does not certify decoding; measure recoverability.